Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 3035
Team ID	SWTID-2026-8933
Project Name	Script-Controlled ACL Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Data Flow Diagrams:

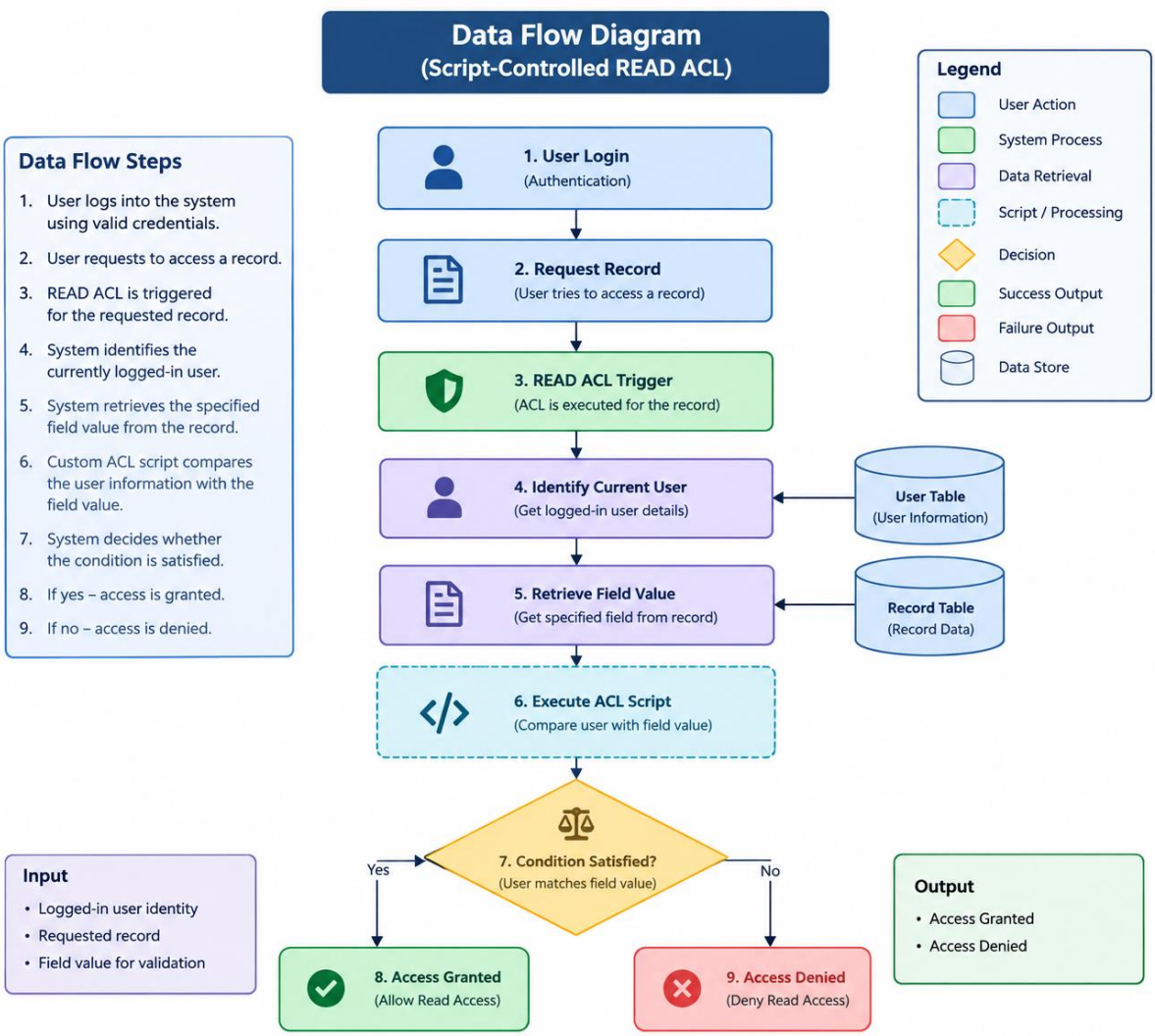
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the rightamount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Acceptance Criteria:

- An authorized user can access the appropriate record.
- An unauthorized user cannot access the restricted record.
- Different field values can be tested.
- The final access behavior matches the defined rules.

Problem statement:

Create a Script-Controlled READ ACL that restricts access to records based on a specific field value. (e.g., only allow access if the user is the assigned user / field value matches certain criteria, otherwise deny access)



User Stories

US-01 — Request Record Access

As a user, I want to access a record so that I can view its information when I have the required permission.

Acceptance Criteria:

- User is authenticated.
- User requests a record.
- READ ACL is triggered for the requested record.

US-02 — Identify Current User

As the system, I want to identify the currently logged-in user so that the user's identity can be checked against the record's field value.

Acceptance Criteria:

- Current user is identified.
- User information is available to the ACL script.
- The information is used for access validation.

US-03 — Validate Field Value

As the system, I want to retrieve the specified field value from a record so that it can be compared with the current user.

Acceptance Criteria:

- Required field is available.
- Field value is retrieved correctly.
- Field value is passed to the validation logic.

US-04 — Allow Authorized Access

As an authorized user, I want to read a record when my identity matches the required field value so that I can access the records assigned to me.

Acceptance Criteria:

- Current user matches the required field value.
- ACL script evaluates the condition as **true**.
- Read access is granted.

US-05 — Restrict Unauthorized Access

As a system administrator, I want to prevent users from reading records when they do not satisfy the defined condition so that restricted records remain protected.

Acceptance Criteria:

- Current user does not match the required field value.
- ACL script evaluates the condition as **false**.
- Read access is denied.

US-06 — Handle Missing Field Values

As the system, I want to handle missing or invalid field values so that access is not accidentally granted.

Acceptance Criteria:

- Missing or invalid values are detected.
- The validation condition is not satisfied.
- Read access is denied.

US-07 — Verify Access Control

As an administrator, I want to test the READ ACL with different users so that I can verify that authorized users can access records and unauthorized users cannot.

Acceptance Criteria:

- Authorized user can read the permitted record.
- Unauthorized user cannot read the restricted record.
- Different field values can be tested.
- Access results match the defined ACL conditions.